

Assignment 03

Registration No. : 2025/CS/079

Index No. : 25000799

Temp. Index No. : 25001078

Activity 01

(01)

a.

```
chamal@ubuntu:/$ cd /
chamal@ubuntu:/$ pwd
/
chamal@ubuntu:/$ ls
bin boot cdrom dev etc home lib lost+found media mnt opt proc root run sbin snap srv swap.img sys tmp usr var
chamal@ubuntu:/$
```

b.

```
chamal@ubuntu:/$ ls -l
total 3985484
lrwxrwxrwx    1 root root          7 Apr 20 14:16 bin -> usr/bin
drwxr-xr-x    5 root root     4096 May 31 18:25 boot
dr-xr-xr-x    2 root root     4096 Apr 22 14:36 cdrom
drwxr-xr-x   20 root root     4140 Jun 12 15:01 dev
drwxr-xr-x  147 root root    12288 Jun  8 08:52 etc
drwxr-xr-x    4 root root     4096 Jun  5 15:20 home
lrwxrwxrwx    1 root root          7 Apr 20 14:16 lib -> usr/lib
drwx-----    2 root root    16384 May 31 16:57 lost+found
drwxr-xr-x    2 root root     4096 Apr 22 14:02 media
drwxr-xr-x    2 root root     4096 Apr 22 14:02 mnt
drwxr-xr-x    3 root root     4096 Jun  5 23:01 opt
dr-xr-xr-x  282 root root          0 Jun 12 15:01 proc
drwx-----    6 root root     4096 Jun  1 15:54 root
drwxr-xr-x   39 root root     1040 Jun 12 15:14 run
lrwxrwxrwx    1 root root          8 Apr 20 14:16 sbin -> usr/sbin
drwxr-xr-x   10 root root     4096 May 31 17:57 snap
drwxr-xr-x    2 root root     4096 Apr 22 14:02 srv
-rw-----    1 root root 4081057792 May 31 17:08 swap.img
dr-xr-xr-x   13 root root          0 Jun 12 15:14 sys
drwxrwxrwt   17 root root      380 Jun 12 15:10 tmp
drwxr-xr-x   11 root root     4096 Apr 22 14:02 usr
drwxr-xr-x   15 root root     4096 Jun  8 08:52 var
chamal@ubuntu:/$
```

C.

```
chamal@ubuntu:/$ ls -lh
total 3.9G
lrwxrwxrwx   1 root root    7 Apr 20 14:16 bin -> usr/bin
drwxr-xr-x   5 root root 4.0K May 31 18:25 boot
dr-xr-xr-x   2 root root 4.0K Apr 22 14:36 cdrom
drwxr-xr-x  20 root root 4.1K Jun 12 15:01 dev
drwxr-xr-x 147 root root 12K Jun  8 08:52 etc
drwxr-xr-x   4 root root 4.0K Jun  5 15:20 home
lrwxrwxrwx   1 root root    7 Apr 20 14:16 lib -> usr/lib
drwx-----  2 root root 16K May 31 16:57 lost+found
drwxr-xr-x   2 root root 4.0K Apr 22 14:02 media
drwxr-xr-x   2 root root 4.0K Apr 22 14:02 mnt
drwxr-xr-x   3 root root 4.0K Jun  5 23:01 opt
dr-xr-xr-x 278 root root    0 Jun 12 15:01 proc
drwx-----  6 root root 4.0K Jun  1 15:54 root
drwxr-xr-x  39 root root 1.1K Jun 12 15:14 run
lrwxrwxrwx   1 root root    8 Apr 20 14:16 sbin -> usr/sbin
drwxr-xr-x  10 root root 4.0K May 31 17:57 snap
drwxr-xr-x   2 root root 4.0K Apr 22 14:02 srv
-rw-----  1 root root 3.9G May 31 17:08 swap.img
dr-xr-xr-x  13 root root    0 Jun 12 15:14 sys
drwxrwxrwt  17 root root 380 Jun 12 15:10 tmp
drwxr-xr-x  11 root root 4.0K Apr 22 14:02 usr
drwxr-xr-x  15 root root 4.0K Jun  8 08:52 var
chamal@ubuntu:/$
```

(02)

a.

```
chamal@ubuntu:/$ ls /etc
ModemManager          cron.weekly           gnome                 ld.so.conf.d         netconfig             rc4.d                 supercat
NetworkManager        cron.yearly           gnome-remote-desktop ldconfig              netplan               rc5.d                 sysctl.d
PackageKit            cronab               gnutls               legal                network              rc6.d                 sysstat
UPower               crypttab             gprofng.rc          libaudit.conf        networkd-dispatcher  rcS.d                 systemd
X11                  cupshelpers          groff               libblockdev          networks             resolv.conf           terminfo
adduser.conf          dbus-1              group               libbiverbs.d         newt                 rmt                   texmf
alsa                 dconf               grub.d              libnl-3              nftables.conf        rpc                   tmpfiles.d
alternatives          debconf.conf        gshadow             libpaper.d           nginx                rsyslog.conf          tpm2-tss
apache2               debian_version       gshadow             libreoffice          nsswitch.conf        sane.d                ts.conf
apparmor             debuginfod          gss                 lighttpd             overlayroot.conf     screenrc              ubuntu-advantage
apparmor.d           default            hdparm.conf        locale.conf          pam.conf             security              ucf.conf
appport              deluser.conf        hosts               locale.gen            pam.d                selinux               udev
apt                  dhcp                host.conf          localtime            paperspecs           sensors.d             udisks2
avahi                dhcpcd.conf         hostname           logrotate.conf       passwd               sensors3.conf         ufw
bash.bashrc          dictionaries-common dpkg                logrotate.d          passwd-              services              update-manager
bash_completion      dracut.conf         hosts              logrotate.d          perl                 sgml                  update-motd.d
bash_completion.d    dracut.conf.d       hosts.allow        lvm                  pki                  shadow                update-notifier
bindresvport.blacklist e2scrub.conf        hosts.deny         magic                 plymouth             shadow-               usb_modeswitch.d
binfmt.d             emacs               init.d             magic.mime            pm                    shadow-               vconsole.conf
bluetooth            ca-certificates     initramfs          manpath.config       polkit-1             shells                vdpau_wrapper.cfg
ca-certificates.conf chatscripts          initramfs-tools    mdadm                 pm                     skel                  vim
ca-certificates      chrony              environment.d       mke2fs.conf          profile              snmp                  vmware-tools
chatscripts          cloud               flash-kernel       modules               protocols            sos                   vtrgb
colord               console-setup       fonts              modprobe.d            python3              ssh                   vulkan
cracklib             console-setup       fstab              modules-load.d        python3.14           ssl                   wgetrc
credstore            credstore.encrypted gdm3               multipath.conf        rc1.d                subgid                whoopsie
cron.d               cron.daily          geoclue            nanorc                rc2.d                subuid                wpa_supplicant
cron.hourly          cron.monthly        ghostscript        needrestart           rc3.d                sudo.conf             xattr.conf
cron.monthly         gavl.conf           glvnd              issue.net             rcS.d                sudo_logsrvd.conf    xdg
chamal@ubuntu:/$
```

b.

```
chamal@ubuntu:/$ ls /bin /sbin
/bin:
3cpio
GET
HEAD
POST
VGAuthService
X11
Xwayland
'['
aa-enabled
aa-exec
aa-features-abi
aarch64-linux-gnu-addr2line
aarch64-linux-gnu-ar
aarch64-linux-gnu-as
aarch64-linux-gnu-c++filt
aarch64-linux-gnu-cpp
aarch64-linux-gnu-cpp-15
aarch64-linux-gnu-elfedit
aarch64-linux-gnu-gcc
aarch64-linux-gnu-gcc-15
aarch64-linux-gnu-gcc-ar
aarch64-linux-gnu-gcc-ar-15
aarch64-linux-gnu-gcc-nm
aarch64-linux-gnu-gcc-nm-15
aarch64-linux-gnu-gcc-ranlib
aarch64-linux-gnu-gcc-ranlib-15
aarch64-linux-gnu-gcov
aarch64-linux-gnu-gcov-15
aarch64-linux-gnu-gcov-dump
aarch64-linux-gnu-gcov-dump-15
aarch64-linux-gnu-gcov-tool
aarch64-linux-gnu-gcov-tool-15
aarch64-linux-gnu-gprof
aarch64-linux-gnu-gprof
aarch64-linux-gnu-ld
aarch64-linux-gnu-ld.bfd
aarch64-linux-gnu-lto-dump
aarch64-linux-gnu-lto-dump-15
fadvise
fallocate
false
fc-cache
fc-cat
fc-conflist
fc-list
fc-match
fc-pattern
fc-query
fc-scan
fc-validate
fdtdump
fdtget
fdtoverlay
fdtput
fgconsole
fgrep
file
finalrd
find
findcore
findmnt
firefox
flock
fmt
fmtutil
fmtutil-sys
fmtutil-user
fold
fonttoolsfont
free
ftp
funzip
fuser
fusermount
fusermount3
ipcrm
ipcs
iptables-xml
ischroot
iscsiadm
isosize
ispell-wrapper
join
journalctl
jq
json-patch-jsondiff
json_pp
jsondiff
jsonpatch
jsonpointer
jsonschema
kbd_mode
kbbinfo
kbbutil
kernel-install
kill
killall
kmod
kmodsign
kpseaccess
kpsereadlink
kpsestat
kpsewhich
kwboot
l2ping
l2test
landscape-sysinfo
latex
ld
ld.bfd
ld.so
ldattach
pktype
pl2pm
pldd
plog
pltotf
plymouth
pmap
pmmap
pmmap
pmmap
pod2html
pod2man
pod2text
pod2usage
podchecker
poff
pollinate
pon
pooltype
powerprofilesctl
ppltotf
pr
precat
preconv
prepmx
preunzip
prezip
prezip-bin
printenv
printf
prlimit
pro
prove
prtstat
ps
ps2pk
psfaddtable
psfgettable
sudo
sudo-rs
sudo-ws
sudoedit
sudoedit-rs
sudoreplay.ws
sum
switch_root
switcherooctl
sync
syncex
systemctl
systemd-ac-power
systemd-analyze
systemd-ask-password
systemd-cat
systemd-cgls
systemd-cgtop
systemd-confext
systemd-creds
systemd-cryptenroll
systemd-cryptsetup
systemd-delta
systemd-detect-virt
systemd-escape
systemd-firstboot
systemd-hwdb
systemd-id128
systemd-inhibit
systemd-machine-id-setup
systemd-mount
systemd-mute-console
systemd-notify
systemd-path
systemd-pty-forward
systemd-run
systemd-socket-activate
```

c.

```
chamal@ubuntu:/$ ls ~
Desktop Documents Downloads Music Pictures Public Templates Videos assignment main.c snap
chamal@ubuntu:/$
```

(03)

```
chamal@ubuntu:/$ ls -la ~
. .bash_history .bashrc .config .local .ssh Desktop Downloads Pictures Templates assignment snap
.. .bash_logout .cache .icons .profile .viminfo Documents Music Public Videos main.c
chamal@ubuntu:/$
```

(04)

```
chamal@ubuntu:/$ mkdir -p ~/practice/dir1/dir2/dir3
chamal@ubuntu:/$
```

(05)

```
chamal@ubuntu:/$ cd /etc
chamal@ubuntu:/etc$ pwd
/etc
chamal@ubuntu:/etc$ mkdir ~/newdir
chamal@ubuntu:/etc$
```

(06)

```
chamal@ubuntu:/etc$ rmdir ~/practice
rmdir: failed to remove '/home/chamal/practice': Directory not empty
chamal@ubuntu:/etc$
```

It throws the above error. It happens because we made sub folders like dir1 inside the practice folder. Since the directory is not empty the system won't let me delete it.

```
chamal@ubuntu:/etc$ rm -rf ~/practice
chamal@ubuntu:/etc$
```

(07)

```
chamal@ubuntu:/etc$ pushd /usr
/usr /etc
chamal@ubuntu:/usr$ pushd /var/log
/var/log /usr /etc
chamal@ubuntu:/var/log$ popd
/usr /etc
chamal@ubuntu:/usr$ popd
/etc
chamal@ubuntu:/etc$
```

```
pushd [-n] [+n] [-n]
pushd [-n] [dir]
Add a directory to the top of the directory stack, or rotate the stack, making the new top of the stack the current working directory. With no arguments, pushd exchanges the top two elements of the directory stack. Arguments, if supplied, have the following meanings:
-n Suppress the normal change of directory when rotating or adding directories to the stack, only manipulate the stack.
+n Rotate the stack so that the nth directory (counting from the left of the list shown by dirs, starting with zero) is at the top.
-n Rotates the stack so that the nth directory (counting from the right of the list shown by dirs, starting with zero) is at the top.
dir Adds dir to the directory stack at the top.

After the stack has been modified, if the -n option was not supplied, pushd uses the cd builtin to change to the directory at the top of the stack. If the cd fails, pushd returns a non-zero value.

Otherwise, if no arguments are supplied, pushd returns zero unless the directory stack is empty. When rotating the directory stack, pushd returns zero unless the directory stack is empty or n specifies a non-existent directory stack element.

If the pushd command is successful, bash runs dirs to show the final contents of the directory stack.
```

```
popd [-n] [+n] [-n]
Remove entries from the directory stack. The elements are numbered from 0 starting at the first directory listed by dirs, so popd is equivalent to "popd +0." With no arguments, popd removes the top directory from the stack, and changes to the new top directory. Arguments, if supplied, have the following meanings:
-n Suppress the normal change of directory when removing directories from the stack, only manipulate the stack.
+n Remove the nth entry counting from the left of the list shown by dirs, starting with zero, from the stack. For example: "popd +0" removes the first directory, "popd +1" the second.
-n Remove the nth entry counting from the right of the list shown by dirs, starting with zero. For example: "popd -0" removes the last directory, "popd -1" the next to last.

If the top element of the directory stack is modified, and the -n option was not supplied, popd uses the cd builtin to change to the directory at the top of the stack. If the cd fails, popd returns a non-zero value.

Otherwise, popd returns false if an invalid option is supplied, the directory stack is empty, or n specifies a non-existent directory stack entry.

If the popd command is successful, bash runs dirs to show the final contents of the directory stack, and the return status is 0.
```

Pushd adds folders to the directory stack while changing the current directory and popd remove the current directory and move to the next directory in the directory stack.

(08)

a.

```
chamal@ubuntu:/etc$ file /bin/cat
/bin/cat: symbolic link to ../lib/cargo/bin/coreutils/cat
chamal@ubuntu:/etc$
```

b.

```
chamal@ubuntu:/etc$ file /etc/passwd
/etc/passwd: ASCII text
chamal@ubuntu:/etc$
```

c.

```
chamal@ubuntu:/etc$ file /usr/bin/passwd
/usr/bin/passwd: setuid ELF 64-bit LSB pie executable, ARM aarch64, version 1 (SYSV), dynamically linked, interpreter /lib/ld-linux-aarch64.so.1, BuildID[sha1]=e951f71c3919970542ae63dea23ec2323d8c23b, for GNU/Linux 3.7.0, stripped
chamal@ubuntu:/etc$
```

Activity 02

(01)

a.

```
chamal@ubuntu:/etc$ file /bin/ls
/bin/ls: symbolic link to ../lib/cargo/bin/coreutils/ls
chamal@ubuntu:/etc$
```

b.

```
chamal@ubuntu:/etc$ file /etc/hosts
/etc/hosts: ASCII text
chamal@ubuntu:/etc$
```

c.

```
chamal@ubuntu:/etc$ file /usr/bin/less
/usr/bin/less: ELF 64-bit LSB pie executable, ARM aarch64, version 1 (SYSV), dynamically linked, interpreter /lib/ld-linux-aarch64.so.1, BuildID[sha1]=b43a3c7da8ab968eaa4bd7f10892896bacc9dde6, for GNU/Linux 3.7.0, stripped
chamal@ubuntu:/etc$
```

(02)

```
chamal@ubuntu:/etc$ mkdir ~/demo_lab && cd ~/demo_lab
chamal@ubuntu:~/demo_lab$
```

(03)

```
chamal@ubuntu:~/demo_lab$ touch alpha.txt beta.txt
chamal@ubuntu:~/demo_lab$
```

(04)

```
chamal@ubuntu:~/demo_lab$ touch -t 2606100340 beta.txt
chamal@ubuntu:~/demo_lab$ ll
total 8
drwxrwxr-x  2 chamal chamal 4096 Jun 12 15:37 ./
drwxr-x--- 19 chamal chamal 4096 Jun 12 15:37 ../
-rw-rw-r--  1 chamal chamal    0 Jun 12 15:37 alpha.txt
-rw-rw-r--  1 chamal chamal    0 Jun 10 03:40 beta.txt
chamal@ubuntu:~/demo_lab$
```

(05)

```
chamal@ubuntu:~/demo_lab$ cp beta.txt backup_beta.txt
chamal@ubuntu:~/demo_lab$ ll
total 8
drwxrwxr-x  2 chamal chamal 4096 Jun 12 15:41 ./
drwxr-x--- 19 chamal chamal 4096 Jun 12 15:37 ../
-rw-rw-r--  1 chamal chamal    0 Jun 12 15:37 alpha.txt
-rw-rw-r--  1 chamal chamal    0 Jun 12 15:41 backup_beta.txt
-rw-rw-r--  1 chamal chamal    0 Jun 10 03:40 beta.txt
chamal@ubuntu:~/demo_lab$
```

(06)

```
chamal@ubuntu:~/demo_lab$ mv alpha.txt alpha_data.txt
chamal@ubuntu:~/demo_lab$
```

(07)

```
chamal@ubuntu:~/demo_lab$ ls -lah
total 8.0K
drwxrwxr-x  2 chamal chamal 4.0K Jun 12 15:41 .
drwxr-x--- 19 chamal chamal 4.0K Jun 12 15:37 ../
-rw-rw-r--  1 chamal chamal    0 Jun 12 15:37 alpha_data.txt
-rw-rw-r--  1 chamal chamal    0 Jun 12 15:41 backup_beta.txt
-rw-rw-r--  1 chamal chamal    0 Jun 10 03:40 beta.txt
chamal@ubuntu:~/demo_lab$
```

(08)

```
chamal@ubuntu:~/demo_lab$ cat > logfile.txt
a. Pgsql
b. Copy
c. Edit
d. Login
e. Logout
f. Restart
g. Shutdown
h. Sleep
chamal@ubuntu:~/demo_lab$
```

(09)

```
chamal@ubuntu:~/demo_lab$ tac logfile.txt
h. Sleep
g. Shutdown
f. Restart
e. Logout
d. Login
c. Edit
b. Copy
a. Pgsql
chamal@ubuntu:~/demo_lab$
```

(10)

```
chamal@ubuntu:~/demo_lab$ echo -e 'Bash\ncopy\nEdit\nLinux\\\\Training\necho "This is quoted text"'
Bash
copy
Edit
Linux\\Training
echo "This is quoted text"
chamal@ubuntu:~/demo_lab$
```

(11)

```
chamal@ubuntu:~/demo_lab$ alias hello='echo Hello from Chamal!'
chamal@ubuntu:~/demo_lab$ hello
Hello from Chamal!
chamal@ubuntu:~/demo_lab$
```

(12)

```
chamal@ubuntu:~/demo_lab$ alias
alias alert='notify-send --urgency=low -i "${ $? = 0 } && echo terminal || echo error)" "$(history|tail -n1|sed -e '\''s/^[0-9]\+\s*//;s/[:&]\s*alert$//'\''\`''"'
alias egrep='egrep --color=auto'
alias fgrep='fgrep --color=auto'
alias grep='grep --color=auto'
alias hello='echo Hello from Chamal!'
alias l='ls -CF'
alias la='ls -A'
alias ll='ls -alF'
alias ls='ls --color=auto'
alias rm='rm -v'
chamal@ubuntu:~/demo_lab$
```

```
chamal@ubuntu:~/demo_lab$ unalias rm
chamal@ubuntu:~/demo_lab$ alias
alias alert='notify-send --urgency=low -i "${ $? = 0 } && echo terminal || echo error)" "$(history|tail -n1|sed -e '\''s/^[0-9]\+\s*//;s/[:&]\s*alert$//'\''\`''"'
alias egrep='egrep --color=auto'
alias fgrep='fgrep --color=auto'
alias grep='grep --color=auto'
alias hello='echo Hello from Chamal!'
alias l='ls -CF'
alias la='ls -A'
alias ll='ls -alF'
alias ls='ls --color=auto'
chamal@ubuntu:~/demo_lab$
```

(13)

```
chamal@ubuntu:~/demo_lab$ set -x
++ __vte_prompt_command
++ local errsv=0
++ __vte_termprop_set vte.shell.postexec 0
++ local errsv=0
++ printf '\033]666;%=s=%s\033\\' vte.shell.postexec 0
++ return 0
++ __vte_osc7
++ local errsv=0
+++ /usr/libexec/vte-urlencode-cwd
++ printf '\033]7;file:///%s%s\033\\' ubuntu /home/chamal/demo_lab
++ return 0
++ local 'pwd=~'
++ '[' /home/chamal/demo_lab '!=' /home/chamal ']'
++ pwd='~/demo_lab'
++ pwd='~/demo_lab'
++ printf '\033]0;%=s@%s:%s\033\\' chamal ubuntu '~/demo_lab'
++ __vte_termprop_signal vte.shell.precmd
++ local errsv=0
++ printf '\033]666;%=s!\033\\' vte.shell.precmd
++ return 0
++ return 0
```



```

chamal@ubuntu:~/demo_lab$ echo hello world
+ echo hello world
hello world
++ __vte_prompt_command
++ local errsv=0
++ __vte_termprop_set vte.shell.postexec 0
++ local errsv=0
++ printf '\033]666;%s=%s\033\\' vte.shell.postexec 0
++ return 0
++ __vte_osc7
++ local errsv=0
+++ /usr/libexec/vte-urlencode-cwd
++ printf '\033]7;file://%s%s\033\' ubuntu /home/chamal/demo_lab
++ return 0
++ local 'pwd=~'
++ '[' /home/chamal/demo_lab '!=' /home/chamal ']'
++ pwd='~/demo_lab'
++ pwd='~/demo_lab'
++ printf '\033]0;%s@%s:%s\033\' chamal ubuntu '~/demo_lab'
++ __vte_termprop_signal vte.shell.precmd
++ local errsv=0
++ printf '\033]666;%s!\033\\' vte.shell.precmd
++ return 0
++ return 0

```

```

chamal@ubuntu:~/demo_lab$ echo $HOME
+ echo /home/chamal
/home/chamal
++ __vte_prompt_command
++ local errsv=0
++ __vte_termprop_set vte.shell.postexec 0
++ local errsv=0
++ printf '\033]666;%s=%s\033\\' vte.shell.postexec 0
++ return 0
++ __vte_osc7
++ local errsv=0
+++ /usr/libexec/vte-urlencode-cwd
++ printf '\033]7;file://%s%s\033\' ubuntu /home/chamal/demo_lab
++ return 0
++ local 'pwd=~'
++ '[' /home/chamal/demo_lab '!=' /home/chamal ']'
++ pwd='~/demo_lab'
++ pwd='~/demo_lab'
++ printf '\033]0;%s@%s:%s\033\' chamal ubuntu '~/demo_lab'
++ __vte_termprop_signal vte.shell.precmd
++ local errsv=0
++ printf '\033]666;%s!\033\\' vte.shell.precmd
++ return 0
++ return 0

```

```
chamal@ubuntu:~/demo_lab$ set +x
+ set +x
chamal@ubuntu:~/demo_lab$
```

(14)

```
chamal@ubuntu:~/demo_lab$ mkdir labtest && echo 'Created Successfully' || echo 'Failed to Create'
Created Successfully
chamal@ubuntu:~/demo_lab$
```

(15)

```
chamal@ubuntu:~/demo_lab$ echo -e 'a. Copy\nb. Edit\nc. 4+4 =8'
a. Copy
b. Edit
c. 4+4 =8
chamal@ubuntu:~/demo_lab$
```